

COMP5046 26s1 Final Practice

USYD 各科往届题库



W1 Introduction and Word Vectors

Q1. MCQ Semester 1 2023 Final (modified)

Select all that apply. Which statements about sparse bag-of-words representations are true? (A) They can store only observed terms. (B) They preserve word order. (C) They are often suitable for fast topic baselines. (D) They require every vocabulary item to be stored for every document.

Q2. Short Answer

Explain why IDF can make a frequent word less useful for classification. Give one concrete example.

Q3. True/False

A Word2Vec cosine similarity of 0.9 proves that two words are interchangeable in every sentence. State true/false and justify.

Q4. Calculation

A corpus has 1000 documents. The term 'enzyme' appears in 10 documents and occurs 4 times in document d. Using $tf = 1 + \log_{10} c$ and $idf = \log_{10}(N/df)$, compute the TF-IDF weight.

Q5. Short Answer

Contrast one-hot, Word2Vec, and contextual embeddings for the word 'bank'.

Q6. MCQ

Which design is most robust to unseen morphology? (A) Word one-hot only. (B) Character/subword features. (C) Removing all rare words. (D) A label majority baseline.

Q7. Long Question

A news classifier using TF-IDF confuses technology policy articles with politics articles. Propose two representation or feature changes and explain the mechanism.

Q8. Table/Diagram

Fill a three-row comparison table for BoW, TF-IDF, and contextual embeddings: information kept, main strength, main failure.

Q9. Short Answer

Why can antonyms sometimes have high distributional similarity?

Q10. MCQ

Skip-gram with negative sampling primarily trains by: (A) predicting nearby context words from a center word. (B) parsing syntax trees. (C) comparing all documents with BLEU. (D) running beam search.



W2 Foundation of NLP Systems

Q1. Short Answer Semester 1 2024 Final (modified)

A spam filter has precision 0.99 but recall 0.05 for spam. Explain why this may still be a poor system.

Q2. MCQ

For named entity recognition on rare entities, which metric exposes minority failure better than accuracy? (A) Macro F1. (B) Raw accuracy only. (C) Training loss only. (D) Vocabulary size.

Q3. Calculation

For a positive class, TP=18, FP=6, FN=12. Compute precision, recall, and F1.

Q4. True/False

Accuracy is always a reliable metric under severe class imbalance. Justify.

Q5. Long Question

Design an error analysis plan for a sentiment model that fails on sarcasm and domain-specific slang.

Q6. Short Answer

What is data leakage? Give one NLP-specific example.

Q7. MCQ

A baseline is useful because it: (A) calibrates whether a complex model helps. (B) proves the task is solved. (C) removes the need for test sets. (D) guarantees fairness.

Q8. Table/Diagram

Map three deployment goals to metrics: avoid false alarms, find all positives, balance both.

Q9. Short Answer

Explain the difference between evaluating an LM and evaluating an application that calls an LM.

Q10. MCQ

Which is most likely leakage? (A) Fitting a TF-IDF vocabulary on train+test. (B) Lowercasing train data. (C) Reporting macro F1. (D) Using a dev set.



W3 Models - Non-linear

Q1. Short Answer Semester 1 2024 Final (modified)

A student removes every activation function from a deep network. What class of functions can the network represent?

Q2. MCQ

Averaged perceptron updates occur when: (A) the prediction is wrong. (B) BLEU decreases. (C) an annotator disagrees. (D) the beam is empty.

Q3. True/False

Adding hidden layers without non-linearities necessarily creates non-linear decision boundaries.

Q4. Calculation

For logits $[2, 1, 0]$, write the softmax probability of class 1 as an expression and approximate it.

Q5. Long Question

A classifier has high train accuracy and low dev F1. Diagnose three likely causes and propose tests.

Q6. Short Answer

Explain why ReLU can create sparse hidden activations and one risk of dead units.

Q7. MCQ

Which intervention most directly addresses overfitting? (A) Early stopping. (B) Evaluating on train only. (C) Duplicating dev into train. (D) Removing the baseline.

Q8. Table/Diagram

Compare linear model, MLP, and contextual encoder for text classification: capacity, cost, interpretability.

Q9. Short Answer

How can feature engineering fix a confusion between Technology and Politics news?

Q10. MCQ

Cross-entropy for the correct class mainly rewards: (A) increasing probability assigned to that class. (B) sorting documents alphabetically. (C) lowering all logits equally. (D) removing labels.



W4 Inference - Greedy and Search

Q1. Short Answer

Explain why beam search is approximate for sequence generation.

Q2. MCQ

Top-p sampling differs from top-k because: (A) candidate set size varies with cumulative probability. (B) it is exact DP. (C) it only works for RNNs. (D) it ignores probabilities.

Q3. Calculation

A beam contains A:-1.0 and B:-1.3. A expands with log probs -0.2,-1.0; B expands with -0.1,-0.4. For beam size 2, list kept scores.

Q4. True/False

Viterbi can only be used with RNNs. Justify.

Q5. Long Question

Given a sequence tagger that outputs illegal BIO transitions, propose an inference-time fix and a training/data fix.

Q6. Short Answer

Contrast CKY parsing and Viterbi tagging in one paragraph.

Q7. MCQ

Increasing beam size usually increases: (A) search cost. (B) guaranteed diversity. (C) annotation agreement. (D) TF-IDF IDF.

Q8. Table/Diagram

Draw or describe the beam update loop: initialise, expand, score, prune, stop.

Q9. Short Answer

Why can length normalisation be needed in generation search?

Q10. MCQ

Greedy decoding can fail because: (A) the locally best prefix may block a better later sequence. (B) it computes all paths. (C) it always samples randomly. (D) it uses gold tokens at test time.



W5 Models - Encoder-decoder

Q1. Short Answer Semester 1 2020 Final (modified)

Where do decoder inputs come from during training and during generation in an encoder-decoder model?

Q2. MCQ

Teacher forcing means: (A) feeding gold previous tokens during training. (B) training without labels. (C) using IDF in decoding. (D) deleting the encoder.

Q3. True/False

A fixed-vector encoder bottleneck can make long-source translation harder.

Q4. Long Question

Design a low-resource machine translation approach using lecture concepts. Include data, model, inference, and evaluation.

Q5. Short Answer

Explain the role of attention in overcoming the fixed-vector bottleneck.

Q6. Calculation

A decoder attention distribution over three source states is $[0.2, 0.5, 0.3]$ and scalar values are $[1, 4, 2]$. Compute the context scalar.

Q7. MCQ

BiRNNs are especially useful for: (A) tasks where a token label benefits from left and right context. (B) hiding future tokens in generation. (C) computing IDF. (D) avoiding all parameters.

Q8. Table/Diagram

Compare word, character, and subword tokenisation for OOV robustness and sequence length.

Q9. Short Answer

State one limitation of BLEU for evaluating translation.

Q10. MCQ

LSTM gates mainly help by: (A) controlling information flow over time. (B) removing recurrence. (C) replacing all embeddings with one-hot. (D) computing kappa.



W6 Models - Transformer

Q1. Short Answer Semester 1 2022 Final (modified)

What is the role of the scaling factor $\sqrt{d_k}$ in scaled dot-product attention?

Q2. MCQ

In self-attention, value vectors primarily supply: (A) the content being mixed. (B) the class imbalance correction. (C) the chance agreement. (D) the IDF denominator.

Q3. Calculation

For query $q=[1,0]$, keys $k_1=[2,0]$, $k_2=[0,1]$, unscaled dot scores are what? Which key gets higher attention before softmax?

Q4. True/False

If all key vectors are identical and masks/position biases are ignored, attention can still distinguish positions by key similarity.

Q5. Long Question

Explain residual stream, multi-head attention, and feed-forward layers as complementary parts of a Transformer block.

Q6. Short Answer

What does RoPE encode, at a high level?

Q7. MCQ

A causal mask prevents: (A) attending to future tokens. (B) using embeddings. (C) all gradient updates. (D) tokenisation.

Q8. Table/Diagram

List Q, K, V, mask, and FFN with one-line roles.

Q9. Short Answer

Why can multi-head attention be more expressive than one head with the same attention pattern?

Q10. MCQ

Layer normalisation mainly helps by: (A) stabilising activation scale. (B) computing BLEU. (C) creating labels. (D) removing softmax.



W7 Models - Large Language Models

Q1. Short Answer

What two objects are compared during vector retrieval in a basic RAG system?

Q2. MCQ

Masked language modelling is most associated with: (A) bidirectional encoders such as BERT. (B) left-to-right GPT generation only. (C) TF-IDF. (D) Cohen kappa.

Q3. Calculation

A 4-token sequence has probability $1/16$ under a language model. Compute perplexity.

Q4. True/False

In-context learning updates the model weights during the prompt.

Q5. Long Question

A RAG answer is fluent but unsupported by retrieved text. Diagnose retrieval, prompting, and generation causes.

Q6. Short Answer

Explain why prompt example order can affect an LLM answer.

Q7. MCQ

Which can reduce inference memory? (A) Quantisation. (B) Increasing precision. (C) Duplicating KV cache. (D) Larger batch with no memory change.

Q8. Table/Diagram

Compare next-token prediction, masked LM, instruction tuning, and preference tuning.

Q9. Short Answer

Why is evaluating an LLM application broader than evaluating perplexity?

Q10. MCQ

Benchmark contamination is a risk because it: (A) inflates apparent generalisation. (B) makes IDF zero. (C) guarantees low latency. (D) removes bias.



W8 Data - Annotation and Crowdsourcing

Q1. Short Answer

State two reasons Cohen's kappa may be low even when annotators are trying carefully.

Q2. MCQ

Select all that apply. Good annotation guidelines should include: (A) label definitions. (B) edge cases. (C) examples. (D) secret labels not shown to annotators.

Q3. Calculation

Two annotators agree on 80 of 100 items. Chance agreement is 0.5. Compute kappa.

Q4. True/False

Model-assisted annotation guarantees that social bias disappears.

Q5. Long Question

Design an annotation quality loop for a 7-point sentiment scale.

Q6. Short Answer

Why can class imbalance complicate interpretation of agreement and accuracy?

Q7. MCQ

Select all that apply. Crowdsourcing quality control may use: (A) gold questions. (B) redundancy. (C) adjudication. (D) no guidelines.

Q8. Table/Diagram

Map ambiguity, vague guidelines, fatigue, and model bias to one mitigation each.

Q9. Short Answer

What is one ethical issue in crowdsourced NLP data collection?

Q10. MCQ

If a model drafts annotations for humans to correct, the main risk is: (A) inherited systematic errors. (B) impossible storage. (C) no time saving ever. (D) automatic perfect kappa.



W9 Training - Unsupervised

Q1. Short Answer Semester 1 2022 Final (modified)

What is masked language modelling and why is it useful for bidirectional encoders?

Q2. MCQ

Token edit detection trains a model to: (A) identify edited versus unchanged tokens. (B) compute Cohen kappa. (C) run CKY only. (D) remove all context.

Q3. True/False

Unsupervised objectives contain no assumptions about what patterns matter.

Q4. Long Question

Compare next-token prediction, MLM, and contrastive learning for representation quality and downstream risks.

Q5. Short Answer

What goes wrong if almost every token is edited in token edit detection?

Q6. Calculation

A clustering run after standardisation has objective 20 and before standardisation has 100. Can you conclude the standardised clustering is better? Explain.

Q7. MCQ

A topic model or clustering output should be checked by: (A) coherence and downstream utility. (B) training objective alone. (C) alphabetic label order. (D) kappa only.

Q8. Table/Diagram

List objective, signal source, and risk for MLM, next-token prediction, contrastive learning.

Q9. Short Answer

Why can self-supervised pretraining still learn social bias?

Q10. MCQ

Domain-adaptive pretraining can help because it: (A) adapts representations to in-domain text. (B) deletes all labels. (C) guarantees fairness. (D) removes evaluation.



W10 Models - Agents

Q1. Short Answer

Define an LLM agent in terms of planning, tool use, observations, and final response.

Q2. MCQ

A tool schema helps by: (A) constraining arguments and outputs. (B) proving the answer is true. (C) removing evaluation. (D) hiding observations.

Q3. True/False

Retrieved web text should be treated as instructions with the same priority as system instructions.

Q4. Long Question

Design an agent to answer course policy questions using retrieval and a calculator/tool. Include safeguards.

Q5. Short Answer

Name two failure modes of tool-using agents.

Q6. Calculation

An agent makes 5 tool calls at 1.2 seconds each plus 3 seconds generation. Estimate latency before network variance.

Q7. MCQ

Select all that apply. Prompt injection mitigation includes: (A) separating data from instructions. (B) allowlisting tools. (C) validating outputs. (D) obeying every retrieved sentence.

Q8. Table/Diagram

Sketch the loop: goal, plan, action, observation, check, final.

Q9. Short Answer

Why is task success a better agent metric than only next-token likelihood?

Q10. MCQ

A stale memory is dangerous because it may: (A) override newer evidence. (B) improve kappa automatically. (C) compute IDF. (D) guarantee grounding.



W11 Training - Reinforcement Learning

Q1. Short Answer

Contrast supervised fine-tuning and preference optimisation.

Q2. MCQ

RLHF usually requires: (A) preference or reward signal. (B) no evaluation. (C) only TF-IDF. (D) a CKY grammar.

Q3. True/False

A reward model perfectly captures all human values once trained.

Q4. Long Question

Explain reward hacking in an LLM assistant and propose two evaluation or training mitigations.

Q5. Short Answer

What role does a KL penalty or reference model play in RL-style LM tuning?

Q6. Calculation

If a policy gets rewards $[1, 0, 1, 1]$ across four episodes, what is the average reward? Why is this insufficient alone?

Q7. MCQ

Select all that apply. Preference data is noisy because: (A) humans disagree. (B) rubrics can be vague. (C) culture and context matter. (D) all labels are objective.

Q8. Table/Diagram

Compare SFT, reward modelling, PPO-style RL, and DPO-style optimisation.

Q9. Short Answer

Why is credit assignment hard in a long tool-using dialogue?

Q10. MCQ

An RL-tuned model becoming overly verbose to win preferences is an example of: (A) reward hacking. (B) IDF smoothing. (C) exact search. (D) kappa correction.



W13 Advanced Topics

Q1. Long Question Semester 1 2020 Final (modified)

Design a QA system for both user-log questions and general web questions. Include routing, retrieval, generation, and evaluation.

Q2. MCQ

For an answer grounded in user logs, the system should prioritise: (A) correct structured retrieval/computation. (B) hallucinated fluent text. (C) BLEU only. (D) ignoring privacy.

Q3. Short Answer

Explain one way coreference resolution can help NER or information extraction.

Q4. True/False

A graph-based dependency parser is always preferred in deployment because accuracy is the only consideration.

Q5. Short Answer

State two ethical concerns for deployed NLP systems.

Q6. Calculation

A system returns 12 relevant answers and 3 irrelevant answers, while missing 8 relevant answers. Compute precision and recall.

Q7. MCQ

Select all that apply. A good short-answer exam response should usually include: (A) claim. (B) mechanism. (C) concrete example. (D) no limitation ever.

Q8. Table/Diagram

Compare dependency parsing, constituency parsing, NER, and coreference by output structure.

Q9. Short Answer

Why can idioms and pragmatic context challenge NLP systems?

Q10. Long Question

A benchmark score improves after adding a large external corpus. Give three checks before claiming real progress.



Answer Key

W1 Introduction and Word Vectors

- Q1. A and C. Sparse representations store non-zero terms and are strong fast baselines, but they lose order and every absent term.
- Q2. IDF downweights terms appearing in many documents, so frequent function words or corpus-wide words such as 'the' or 'movie' may carry little class information.
- Q3. False. Similarity reflects similar contexts, not guaranteed substitution; polysemy, syntax, and antonymy can break interchangeability.
- Q4. $tf = 1 + \log_{10} 4 \approx 1.602$, $idf = \log_{10} 100 = 2$, so weight ≈ 3.204 .
- Q5. One-hot has no similarity and one vector per type; Word2Vec gives a static semantic vector; contextual embeddings can give different vectors for river bank and bank account.
- Q6. B. Subword or character features can represent unseen inflections and compounds.
- Q7. Use phrases/entities such as 'AI regulation' and contextual embeddings or metadata; inspect confusions and add features that separate policy topic from political source/style.
- Q8. BoW: counts/fast/order failure. TF-IDF: rarity/good baseline/no semantics. Contextual embeddings: sense/context/cost and opacity.
- Q9. Antonyms often occur in similar syntactic and topical contexts, e.g. hot/cold near temperature words.
- Q10. A. Skip-gram predicts context from the center word; negative sampling avoids full softmax.

W2 Foundation of NLP Systems

- Q1. It catches only a tiny fraction of spam. High precision alone is weak when false negatives are costly or users still receive nearly all spam.
- Q2. A. Macro F1 averages per label and makes rare-class failures visible.
- Q3. Precision $18/(18 + 6) = 0.75$, recall $18/(18 + 12) = 0.60$, $F1 = 2PR/(P + R) = 0.667$.
- Q4. False. A majority-class classifier can have high accuracy while failing the minority class.
- Q5. Inspect false positives/negatives, slice by slang/domain/length, compare with baseline, add sarcasm examples, check annotation ambiguity, and evaluate macro metrics.
- Q6. Leakage is test information entering training or model selection; e.g. fitting vocabulary/IDF on the whole corpus before splitting.
- Q7. A. A baseline gives a sanity check and a comparison point.
- Q8. Avoid false alarms: precision; find all positives: recall; balance: F1 or task-specific cost.
- Q9. LM evaluation uses perplexity/next-token/benchmarks; application evaluation includes prompting, retrieval, latency, cost, safety, factuality, and user success.
- Q10. A. The test distribution influences the representation.

W3 Models - Non-linear

- Q1. A linear or affine function. Composing linear maps remains linear, so depth alone cannot form nonlinear boundaries.
- Q2. A. If predicted label differs from gold, add features to the gold class and subtract from the predicted class.
- Q3. False. Without nonlinear activations, the stack collapses to one affine map.
- Q4. $e^2/(e^2 + e^1 + e^0) \approx 7.389/(11.107) = 0.665$.
- Q5. Likely overfitting, leakage in train style, label noise, or distribution shift. Test with learning curves, ablations, slice errors, and baseline comparisons.
- Q6. ReLU outputs zero for negative inputs, so many units can be inactive; a unit can die if updates keep it always negative.
- Q7. A. Early stopping limits overfitting to training noise.
- Q8. Linear: low/cheap/interpretable; MLP: medium/moderate/less transparent; contextual encoder: high/expensive/opaque.
- Q9. Add topic entities, phrases, source-independent terms, or contextual embeddings that distinguish AI policy language from party politics.
- Q10. A. Cross-entropy is low when the model assigns high probability to the correct label.

W4 Inference - Greedy and Search

- Q1. It prunes partial hypotheses. A prefix outside the beam may later lead to the global best sequence.
- Q2. A. Top-p chooses the smallest set whose mass reaches threshold p, so set size changes.



- Q3. A paths: -1.2 and -2.0. B paths: -1.4 and -1.7. Keep -1.2 and -1.4.
- Q4. False. Viterbi applies to models with transition and emission/local scores, not only RNNs.
- Q5. Constrain decoding transitions with a mask or CRF/Viterbi; improve labels/guidelines or train with structural and more illegal-transition examples.
- Q6. CKY is chart parsing over spans and grammar rules; Viterbi finds a best state path through a sequence model with transitions/emissions.
- Q7. A. More candidates are retained and expanded.
- Q8. Maintain a list of partial hypotheses, expand each by candidate tokens/tags, add log score, sort, keep top beam size, stop on end condition.
- Q9. Summed log probabilities often prefer short sequences; length normalisation reduces that bias.
- Q10. A. Local choices can be globally poor.

W5 Models - Encoder-decoder

- Q1. Training often uses a start token plus gold previous tokens; generation uses a start token and the model's own previous outputs.
- Q2. A. It feeds gold previous outputs into the decoder during training.
- Q3. True. One vector may not preserve all source details, especially for long sequences.
- Q4. Use multilingual pretrained seq2seq, transfer from high-resource pairs, back-translation, lexicons, subword tokenisation, beam search, BLEU/chrF plus human adequacy/fluency and low-resource slice checks.
- Q5. Attention gives each decoding step a weighted view of source states, allowing access to different source positions rather than one fixed vector.
- Q6. $0.2 \cdot 1 + 0.5 \cdot 4 + 0.3 \cdot 2 = 2.8$.
- Q7. A. It uses both prior and later context for token-level decisions.
- Q8. Word: short but OOV; character: robust but long; subword: compromise.
- Q9. BLEU rewards n-gram overlap and may miss semantic equivalence, adequacy, or factual errors.
- Q10. A. Input/forget/output gates regulate memory and gradients.

W6 Models - Transformer

- Q1. It keeps dot-product magnitudes from growing with dimension, preventing overly peaked softmax and weak gradients.
- Q2. A. Values are weighted and summed to form the mixed representation.
- Q3. Scores are 2 and 0, so k1 is higher.
- Q4. False. Equal keys give equal dot products for a fixed query, ignoring masks and biases.
- Q5. The residual stream carries representations through additive updates; heads attend to different relations/subspaces; FFNs apply token-wise nonlinear transformations.
- Q6. RoPE encodes position by rotating query/key components with position-dependent angles so attention can reflect relative position.
- Q7. A. It blocks access to later tokens in autoregressive decoding/training.
- Q8. Q asks, K matches, V supplies content, mask restricts allowed positions, FFN transforms each token representation.
- Q9. Even with the same attention pattern, heads can use separate learned Q/K/V projections, value subspaces, concatenation, and an output projection, so they can carry different transformed content.
- Q10. A. It normalises hidden activations to stabilise training.

W7 Models - Large Language Models

- Q1. The query embedding and stored document/chunk embeddings are compared.
- Q2. A. MLM uses both sides of context and is common for BERT-style encoders.
- Q3. Perplexity = $(1/(1/16))^{1/4} = 16^{1/4} = 2$.
- Q4. False. The model conditions on examples in the prompt; weights are unchanged.
- Q5. Check whether relevant chunks exist and were retrieved, embedding consistency, chunking, prompt instructions to cite/use evidence, and whether the generator ignored or contradicted context.
- Q6. The prompt is sequential context; recency, format, label order, and accidental correlations can bias continuation.
- Q7. A. Lower numerical precision can reduce memory.
- Q8. NTP predicts next token; MLM predicts masked tokens; instruction tuning learns prompt-response behaviour; preference tuning optimises ranked outputs.



- Q9. Applications include retrieval, prompting, tool use, latency, cost, safety, factuality, and user task success.
- Q10. A. The model may have seen similar benchmark items during training.

W8 Data - Annotation and Crowdsourcing

- Q1. Ambiguous items, unclear guidelines, class imbalance, different interpretations, insufficient calibration, or too many fine labels.
- Q2. A, B, and C. Hidden labels are not a guideline.
- Q3. $\kappa = (0.8 - 0.5) / (1 - 0.5) = 0.6$.
- Q4. False. Model bias and systematic errors can propagate.
- Q5. Pilot labels, define scale anchors, add examples/edge cases, train annotators, measure agreement, revise, adjudicate, and audit class/slice distributions.
- Q6. A model or annotator can look accurate by predicting the majority; chance agreement can be high and minority labels may be poor.
- Q7. A, B, and C. Guidelines remain necessary.
- Q8. Ambiguity: edge-case rules; vague guidelines: examples; fatigue: shorter batches; model bias: human audits and slice checks.
- Q9. Fair pay, consent, privacy, exposure to harmful content, demographic bias, and worker wellbeing are valid issues.
- Q10. A. Drafting can anchor humans and propagate model errors.

W9 Training - Unsupervised

- Q1. MLM masks/replaces tokens and predicts them from surrounding context, enabling both left and right context in encoders.
- Q2. A. It predicts which tokens were edited.
- Q3. False. The corruption, context, negatives, and data distribution impose assumptions.
- Q4. NTP is natural for generation but left-context only; MLM learns bidirectional representations but has mask mismatch; contrastive learning shapes similarity but depends on positives/negatives and can encode false negatives.
- Q5. There is too little reliable context, so the model may not learn meaningful detection.
- Q6. No. Scaling changes the geometry and objective scale, so raw objective values are not directly comparable.
- Q7. A. Human coherence and downstream performance matter.
- Q8. MLM: masked tokens/context/mask mismatch. NTP: next token/prefix bias. Contrastive: pairs or augmentations/false negatives.
- Q9. The raw data contains historical and social patterns, and objectives reproduce statistical regularities.
- Q10. A. It shifts representations toward the target domain, though evaluation is still needed.

W10 Models - Agents

- Q1. An agent uses an LM to decide steps, call external tools, observe results, update state, and produce a grounded answer or action.
- Q2. A. It reduces malformed calls, but truth still depends on tool result and reasoning.
- Q3. False. Retrieved text is untrusted data and must not override higher-priority instructions.
- Q4. Use retrieval over official docs, quote evidence, calculator for dates/weights, separate data/instructions, validate sources, handle uncertainty, log tool calls, and evaluate answer correctness and safety.
- Q5. Wrong tool choice, malformed arguments, prompt injection, stale memory, hallucinated observations, loops, or failure to verify tool output.
- Q6. $5 \times 1.2 + 3 = 9$ seconds.
- Q7. A, B, and C. Retrieved instructions should not be blindly obeyed.
- Q8. Goal $\rightarrow$ plan $\rightarrow$ tool/action $\rightarrow$ observation $\rightarrow$ check against goal/evidence $\rightarrow$ final or repeat.
- Q9. Agents are systems; success depends on tool selection, grounding, state, cost, latency, and final correctness, not just fluent tokens.
- Q10. A. Old state can conflict with current evidence.

W11 Training - Reinforcement Learning

- Q1. SFT imitates demonstrations; preference optimisation changes the model to prefer ranked/chosen outputs over rejected ones.



- Q2. A. It needs rewards or preference comparisons.
- Q3. False. Reward models are proxies and can miss values or be exploited.
- Q4. Reward hacking occurs when the model optimises the proxy, e.g. verbosity or sycophancy, rather than usefulness with diverse evals, adversarial tests, KL control, human audits, and better rubrics.
- Q5. It limits drift from the base/reference model, preserving language quality and reducing extreme optimisation.
- Q6. Average reward = $3/4 = 0.75$. It does not reveal safety, distribution shift, hacking, or which actions caused success.
- Q7. A, B, and C. Human preferences are contextual and inconsistent.
- Q8. SFT: demos/imitate; RM: preference pairs/score proxy; PPO: optimise reward with KL; DPO: direct pairwise preference objective.
- Q9. A final outcome may depend on many earlier tokens/tool calls, so the failing action is not obvious.
- Q10. A. The model exploits a preference proxy.

W13 Advanced Topics

- Q1. Route log questions to authenticated database/API computation. Route open questions to retrieval over web/docs, rank evidence, generate cited answers, and evaluate correctness, grounding, privacy, latency, and user success.
- Q2. A. Structured questions need exact data access and computation.
- Q3. Coreference links mentions such as 'Alice' and 'she', enabling entity attributes/events to be merged across sentences.
- Q4. False. Speed, memory, incremental use, and engineering constraints may favour transition-based parsers.
- Q5. Bias/fairness, privacy/consent, hallucination, accessibility, misuse, worker welfare, and accountability are valid.
- Q6. Precision $12/(12 + 3) = 0.8$; recall $12/(12 + 8) = 0.6$.
- Q7. A, B, and C. Limitations often strengthen the answer.
- Q8. Dependency: head-dependent edges; constituency: nested spans/tree; NER: labelled entity spans; coreference: clusters of mentions.
- Q9. Meaning may not be compositional from words alone and may require world knowledge, speaker intent, or discourse context.
- Q10. Check contamination/leakage, slice/domain robustness, significance, annotation consistency, cost/latency, and whether gains transfer to fresh tasks.